Alexander Martin

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EDUCATION

Johns Hopkins University — **Baltimore, Maryland**
Ph.D. in Computer Science — Expected 05/2029
Advisor: Dr. Benjamin Van Durme

University of Rochester — **Rochester, New York**
B.S. in Computer Science; Highest Honors in Research — 05/2024
Advisor: Dr. Aaron Steven White, Dr. Jiebo Luo
Thesis: Human-Centric Event Representations in Documents and Videos

SELECT HONORS & AWARDS

National Science Foundation, **Graduate Research Fellowship** 2024 – 2029
Computing Research Association, **Outstanding Undergraduate Research Award (HM)** 2024
University of Rochester, **Charles L. Newton Prize** 2024

INDUSTRY RESEARCH

Amazon, Fire TV 08/2025 – 08/2026
Researcher; AI2AI Award

- Researched efficient, scalable multimodal conversational retrieval for video content.
- Developed multi-vector index compression, reducing the index size of video collections by ~97% while matching or improving retrieval quality.
- Presented findings regularly to Amazon Fire TV scientists.

RESEARCH EXPERIENCE

Johns Hopkins University 08/2024 – Present
Ph.D. Researcher; Advised by Dr. Benjamin Van Durme

- Research efficient and scalable collaborative omnimodal retrieval and generation: interactive, reviewable systems that move beyond one-shot RAG toward end-to-end research agents.
- Published on retrieval (CVPR, SIGIR), generation (ACL, EMNLP), and evaluation (EMNLP).
- Mentored students to first-author papers (incl. SIGIR); now in Ph.D., research, and MLE roles.

Human Language Technology Center of Excellence 06/2026 – 08/2026
Program Co-Lead, SCALE 2026 Summer Research Workshop

- Co-Organized and ran a 10-week summer research workshop with 50+ participants.
- Led the 11-person generation team on efficient, grounded article generation from multiple videos; mentored its student interns.
- Built the core data and evaluation infrastructure for the workshop's shared task.

University of Rochester 08/2020 – 05/2024
Undergraduate Researcher; Advised by Dr. Jiebo Luo & Dr. Aaron Steven White

- Multimodal generation and video understanding (ACM MM, EMNLP); event extraction and summarization from text (NAACL, EMNLP).

PUBLICATIONS

Selected Publications: Omnimodal Retrieval, Generation, and Evaluation *(* equal contribution)*

[1] **A. Martin**, et al. "*Knowledge Pull Requests for Continual Document Authoring*" (ArXiv 2026)
[2] **A. Martin**, et al. "*Seeing Through the MiRAGE: Evaluating Multimodal Retrieval Augmented Generation*" (EMNLP 2026)
[3] **A. Martin**, et al. "*WikiVideo: Article Generation from Multiple Videos*" (ACL 2026)
[4] H. Qin*, **A. Martin***, et al. "Multi-Vector Index Compression in Any Modality" (SIGIR 2026)
[5] A. Reddy*, **A. Martin***, et al. "*Video-ColBERT: Contextualized Late Interaction for Text-to-Video Retrieval*" (CVPR 2025)

TALKS

Knowledge Intensive Multimodal Retrieval Augmented Generation [talk] — **November 2025**
University of Maryland, Baltimore County; PPR Seminar — Seminar

Discovering and Understanding Events in Videos in Real Time [slides] — **October 2025**
MacroXStudio (macrox.ai) — Talk

On Demand Article Generation of Events in Real Time [slides] — **April 2025**
University of Rochester, Department of Computer Science — Seminar

MARE: Automatic Modality-Agnostic Report Evaluation [slides] — **April 2025**
Eval4Rag Workshop; European Conference on Information Retrieval — Oral

Understanding Events in Multimodal Data Through Question Answering [slides] — **April 2025**
4th Workshop on Processing and Evaluating Event Representations — Talk

FAMuS: Frames Across Multiple Sources [slides] — **April 2024**
3rd Workshop on Processing and Evaluating Event Representations — Talk

Jurassic World Remake [slides] — **October 2023**
Main Conference; ACM Multimedia — Oral

SERVICE

MAGMaR: Workshop on Multimodal Augmented Generation via Multimodal Retrieval
Organizing Committee; *ACL 2026*
Program Committee; *ACL 2025*

Mentorship
Charles Weng (**Initial Placement:** TikTok, Research Scientist)
Dengjia Zhang (**Initial Placement:** University of Pittsburgh, Ph.D. Student)
Hanxiang Qin (**Initial Placement:** Kuaishou, Machine Learning Engineer)
Katherine Guerrerio (**Current:** Johns Hopkins University, B.S. 2027)

Community Service
North American Computational Linguistics Open Competition (NACLO) — 2024 – Present
Competition Organizer
STEM Initiative — 2021 – 2024
Education Mentor
University of Rochester: Residential Life — 2021 – 2024
Resident Advisor

Teaching
Introduction to Artificial Intelligence (University of Rochester) — Spring 2023
Data Structures and Algorithms (University of Rochester) — Spring 2022, Fall 2022
Introduction to Computer Science (University of Rochester) — Fall 2021

OTHER HONORS & AWARDS

University of Rochester, **Senior Research Award** — 2024
University of Rochester, **Research Presentation Grant (x2)** — 2023, 2024
University of Rochester, **Dean's Award in Engineering and Mathematics (x2)** — 2023, 2024
University of Rochester, **River Campus Libraries Dataset Grant** — 2023
University of Rochester, **Residential Life Best Program of the Year** — 2023
University of Rochester, **Make It Happen Grant** — 2022

SKILLS

Python, PyTorch, Hugging Face, FAISS, PLAID, vLLM